

Business Interpretation and Model Recommendation

BI-ready interpretation of model results and practical model selection guidance

Ruturaj Mokashi

Executive decision guide

This document converts the models into business choices. The goal is not to pick the most complex model. The goal is to use the model that matches the target, data risk and decision need.

Business question	Model to start with	Why	Risk to check
Which marketing channels drive Sales?	Multiple Linear Regression	Clear coefficients and easy explanation.	Correlation between channels and outliers.
Which numeric prediction model is stable with related features?	Ridge or ElasticNet	Shrinks unstable coefficients.	Scaling and leakage.
Which features can be removed?	Lasso / LassoCV	Can set coefficients to zero.	Do not interpret zero coefficients blindly if features are correlated.
How many rentals will occur?	Poisson or Negative Binomial GLM	Count-aware model family.	Overdispersion and leakage from cnt components.
Who should be targeted in a campaign?	Logistic Regression	Outputs probabilities and supports threshold/lift decisions.	Imbalance, calibration and threshold choice.

Business interpretation by notebook

Notebook	Best/important result	Business meaning
Notebook 1	Multiple regression $R^2 = 0.8870$, RMSE = 1.6846; TV-only $R^2 = 0.6401$, RMSE = 3.0064.	Using TV, Radio and Newspaper together explains Sales much better than TV alone.
Notebook 2	LassoCV selected $\alpha = 1.6552$ with test RMSE = 52.9196 and Test $R^2 = 0.4714$.	Regularization improved stability and selected a simpler model by shrinking weak features.
Notebook 3	Linear Regression had best MAE = 8.194; Negative Binomial had better count-model deviance = 4.213.	For operational accuracy use the best MAE model; for count-model fit, NB is more statistically suitable when overdispersion exists.
Notebook 4	Baseline Logistic Regression accuracy = 0.777, ROC-AUC = 0.614, PR-AUC = 0.349.	The model ranks responders somewhat better than random but threshold tuning is required for campaign targeting.
Notebook 5	Closed-form OLS matched statsmodels coefficients exactly; campaign multiclass model test accuracy = 0.700.	Diagnostics confirm how coefficients are learned and how to test model assumptions.

Model recommendation table

Model	Best when	Business explanation
Linear Regression	Relationship is mostly linear and interpretability matters.	Use it to explain drivers and predict numeric outcomes.
Ridge	Many features are correlated or coefficients look unstable.	Use it when you want stability without dropping features.
Lasso	You want a simpler model and feature selection.	Use it when removing weak variables is useful.

Model	Best when	Business explanation
ElasticNet	You want shrinkage and selection in correlated data.	Use it when Ridge and Lasso each solve only part of the problem.
Poisson	Target is a count and variance is close to mean.	Use it for count outcomes with simple count assumptions.
Negative Binomial	Count target has overdispersion.	Use it when Poisson underestimates uncertainty.
Logistic Regression	Target is binary or categorical.	Use it for probability scoring and threshold-based decisions.

References and source foundation

The documents use the completed Day 12 notebooks as the main project evidence. The theory explanations are also aligned with the uploaded course materials and official library documentation.

Source	How it was used
Uploaded course guides: leit03.pdf and Haupt_leitfaden03.pdf	OLS, least squares, residuals, regression line, correlation, interpolation/extrapolation.
Uploaded ML slides: Methode der Kleinsten Quadrate and Typen des Machinellen Lernens	Supervised learning, regression vs classification, gradient descent, logistic regression context.
scikit-learn Linear Models User Guide - https://scikit-learn.org/stable/modules/linear_model.html	Linear Regression, Ridge, Lasso, ElasticNet, Logistic Regression, PoissonRegressor.
scikit-learn Model Evaluation User Guide - https://scikit-learn.org/stable/modules/model_evaluation.html	MSE, MAE, R2, ROC-AUC, PR-AUC, precision, recall, F1, calibration.
statsmodels GLM documentation - https://www.statsmodels.org/stable/glm.html	Generalized Linear Models, Poisson family and model interpretation.
statsmodels NegativeBinomial documentation - https://www.statsmodels.org/stable/generated/statsmodels.discrete.discrete_model.NegativeBinomial.html	Negative Binomial implementation details and count-model context.
UCLA IDRE Negative Binomial Regression notes - https://stats.oarc.ucla.edu/r/dae/negative-binomial-regression/	Practical explanation of Negative Binomial as a model for overdispersed count outcomes.